

⚡ GEN AI WORKSHOP PROJECT PRESENTATION

Insightful

AI Powered Customer Feedback Analysis Platform

 Generative AI

 RAG System

 Analytics



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Massive Volume of Customer Feedback



Unstructured & Text-Heavy Data



Manual Analysis is Time-Consuming



Missed Business Insights & Trends

Feedback Data Sources



- App Store Reviews
- Surveys
- Social Media
- Support Chats

💡 "Customer feedback is a goldmine if analyzed correctly."

The Problem

Manual & Unstructured Analysis

-  **Inefficient Manual Analysis**
Reading thousands of reviews manually consumes excessive time and resources.
-  **Lack of Context**
Simple keyword searches miss sarcasm, nuance, and deeper meanings.
-  **No Prioritization**
Critical bugs get buried under minor feature requests and noise.
-  **Hard to Extract Insights**
Converting free text into quantitative business data is difficult.

vs

Insightful Solution

AI-Powered Intelligence

-  **Automated Ingestion**
AI instantly processes feedback from multiple sources in real-time.
-  **Sentiment & Category Detection**
Automatically tags feedback as Positive/Negative and assigns categories.
-  **Semantic Search (RAG)**
Understand user intent ("login broken") beyond simple keywords.
-  **Dashboard Insights**
Visualizes trends and actionable items for immediate decision making.



Core Objectives

- 1 Automate feedback analysis using **Generative AI** to handle large volumes.
- 2 Convert **unstructured** text feedback into structured, queryable data.
- 3 Enable **Natural Language Querying** for non-technical users via RAG.
- 4 Provide **real-time business insights** through an interactive dashboard.



Success Criteria

- ☒ Accurate Sentiment & Category Detection
- ☒ Fast Semantic Search Responses (< 2s)
- ☒ Intuitive & Easy-to-Understand Dashboard
- ☒ Actionable Outputs (e.g., Bug Tickets)



Product Managers

DEVELOPMENT FOCUS

- Identify critical **feature requests** from user noise.
- Prioritize **bug fixes** based on impact and frequency.
- Map product roadmap to actual customer needs.



Marketing Teams

BRAND PERCEPTION

- Track real-time **customer sentiment** trends.
- Automatically extract positive **testimonials** for campaigns.
- Monitor campaign reaction across social channels.



Startup Founders

STRATEGIC VISION

- Understand overall **product perception** instantly.
- Make fast, **data-driven decisions** for pivots.
- Identify churn risks before they impact revenue.



01

Feedback Ingestion & Auto-Categorization

Automatically aggregates unstructured feedback from multiple sources and organizes it into predefined categories.



02

Sentiment Analysis

Classifies feedback as Positive, Negative, or Neutral with high accuracy to gauge overall customer satisfaction.



03

Semantic Search using RAG

Leverages vector embeddings to understand context, enabling natural language queries like "Why are users churning?"



04

Emerging Theme Detection

Identifies rising topics and hidden patterns in feedback before they become critical issues.



05

Actionable Item Generation

Converts raw feedback into structured outputs like bug reports, feature requests, or support tickets.



06

Competitive Feedback Analysis

Benchmarks your product's perception against competitors to identify market gaps and opportunities.

Workflow Case Brief

- 1 User Input**
User pastes feedback or asks: "Why are login attempts failing?"
- 2 Natural Language Query**
System interprets the intent behind the user's question.
- 3 Semantic Retrieval**
Fetches relevant reviews from Pinecone even if exact keywords miss.
- 4 AI Insight Generation**
Gemini summarizes findings into actionable business insights.





Emerging Theme Detection

Proactively identifies rising topics and sudden shifts in customer conversation before they escalate.



Action Item Generator

Automatically converts feedback into structured Jira tickets for bugs or feature requests.



Competitive Snapshot Generator

Analyzes competitor reviews to highlight market gaps and your product's comparative strengths.



AI-Based Feedback Summarization

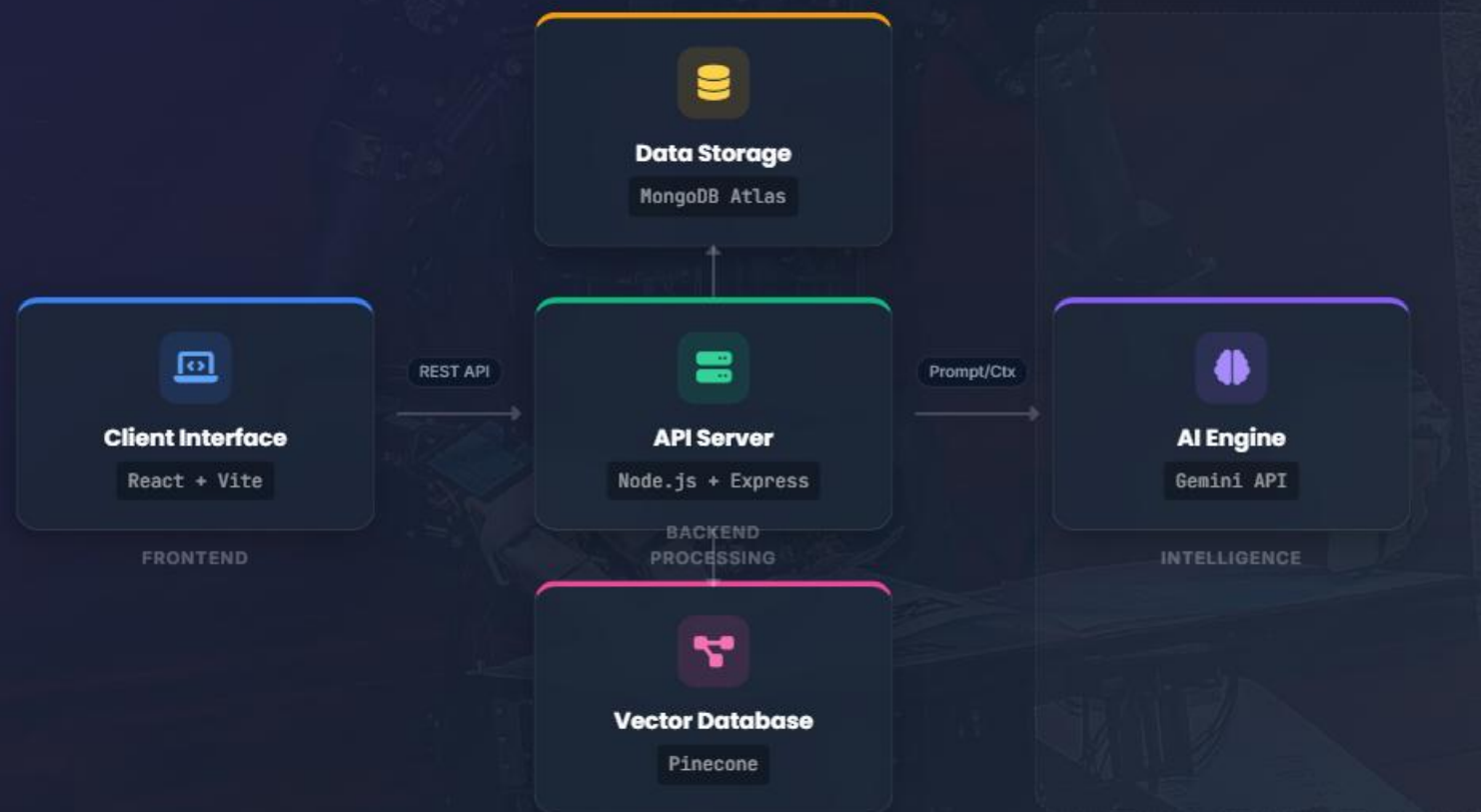
Condenses thousands of reviews into concise executive summaries for weekly reporting.



Trend & Pattern Identification

Detects long-term sentiment shifts and correlations between feature releases and user satisfaction.





PHASE A Data Ingestion & Processing



Feedback Input

Raw text from APIs & uploads



AI Classification

Sentiment & Category tagging



Embedding Gen

Text-to-Vector conversion



Hybrid Storage

MongoDB + Pinecone

PHASE B Retrieval Augmented Generation (RAG)



User Query

Natural Language Question



Semantic Retrieval

Fetch top-k matches



Context Augment

Combine Query + Context



AI Response

Insightful Answer Generation



Bridges the gap between customer voice and strategic business decisions.



AI enables faster, smarter, and context-aware feedback analysis.



Reduces manual effort while significantly improving product quality.



Scalable, full-stack solution designed for modern agile businesses.



Turning customer feedback into actionable intelligence using AI.

Insightful Platform – Final Year Project 2026